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Subject : Database Management system
Btech CSE core

ASSIGNMENT

"Section - A"

Ques-① What is DBMS? Why is it required?
A database management system is a software that is used to store, manage and retrieve data efficiently.
It is required because:
- it reduces data redundancy
- it maintains data consistency
- it provides security
- it supports backup and recovery
- it allows multiple users to access data

Ques-② Why is DBMS better than the traditional file systems?
In a traditional file system, data duplication and inconsistency can occur easily. There is also limited security and no proper backup system.
DBMS is better because it:
- reduces redundancy
- prevents inconsistency
- provides better security
- supports concurrency
- has backup and recovery mechanisms.

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Ques ③ What is Data Abstraction? Explain its levels?
Data abstraction means hiding complex internal details and showing only essential information to users.
It has 3 levels:

(i) physical level -
It describes how data is stored physically.

(ii) logical level -
It describes the structure of the entire database.

(iii) view level -
It describes how users see the data.

Ques ④ What is three-level schema Architecture? Why is it important?
Three-schema architecture divides the database into 3 levels:

→ Internal level
→ Conceptual level
→ External level.

It is important because it provides data independence and separates user views from physical storage.

Ques ⑤ What is data independence? Why is it needed?
Data independence means the ability to change the schema at one level without affecting other levels.

Types:
- physical data independence
- logical data independence.

It is needed for flexibility and easy database modification.

Ques-⑥ What is an ER Model?
The entity-Relationship (ER) model is a conceptual model used to represent real-world entities, their attributes, and relationships in a graphical form.

Ques-⑦ What are integrity-constraints? Why are they important?
Integrity constraints are rules that ensures the accuracy and consistency of data in a database.
Examples-
- Primary key
- Foreign key
- Unique
- Not null
- Check.

They prevent invalid or duplicate data from entering the database.

Section- 'B'

Ques- Draw three-schema architecture and explain each level.

External level
(user-view/Applications) → shows how different user view specific parts of database

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Conceptual level
(Logical database schema) → Describes the overall logical structure of entire database

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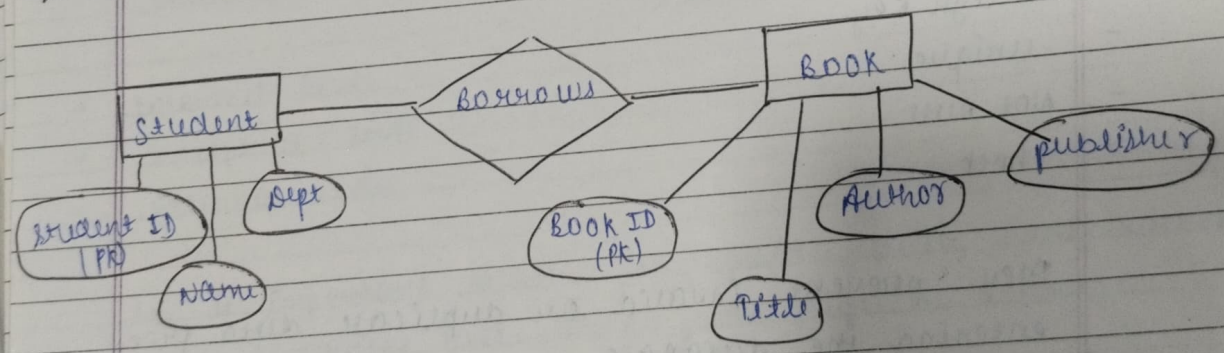
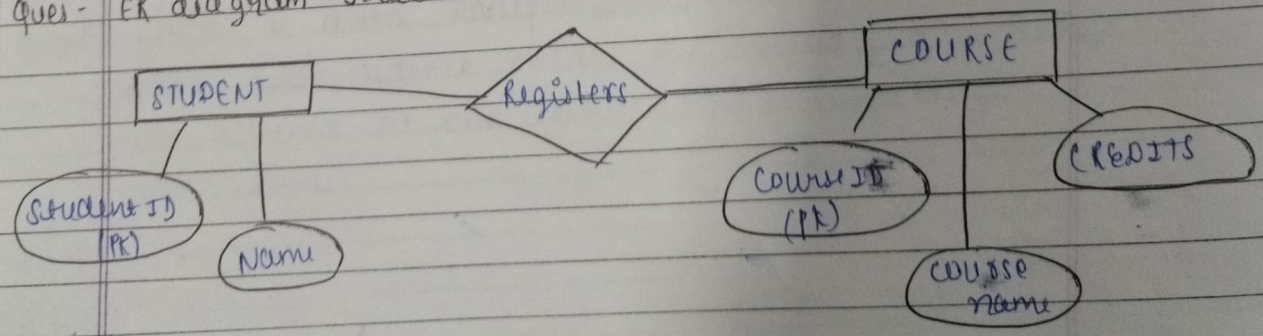
Internal level
(physical storage level) → Describes how data is physically stored in memory.

Ques- ER diagram - Library management system



Ques- ER diagram - Student course Registration



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section - 'C'

Ques- A hospital wants to store patients, doctor, and appointment data. Identify entity, attribute and relationship.

- Entities: patient, doctor, appointment
- Attributes:
  - patient (patient_ID, Name, Age)
  - doctor (Doctor_ID, Name, Specialization)
  - appointment (Appointment_ID, Date, Time)
- Relationship:
  - Doctor treats patients
  - Patient books Appointments.

Ques- An online shopping platform stores customer, product and order data. Suggest keys and constraints.

- primary keys ⇒
  - cust_ID
  - Product_ID
  - order_ID
- Constraints ⇒
  - Foreign key
  - NOT NULL
  - UNIQUE
  - CHECK (price > 0)

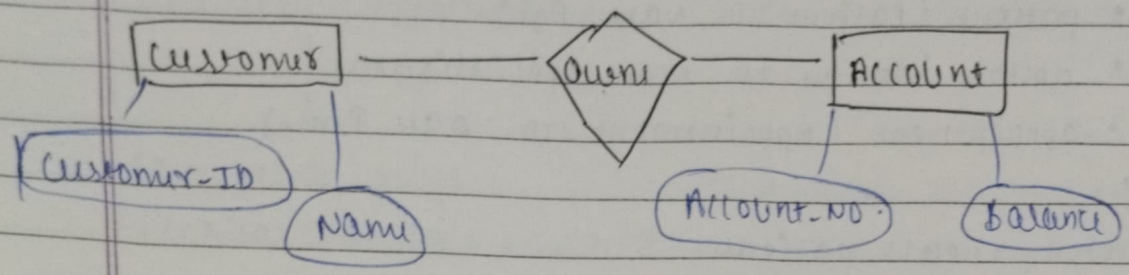
Ques- A university stores student records. Explain how data abstraction helps different users.

- View level - student access
- logical level - Admin management
- physical level - Data storage

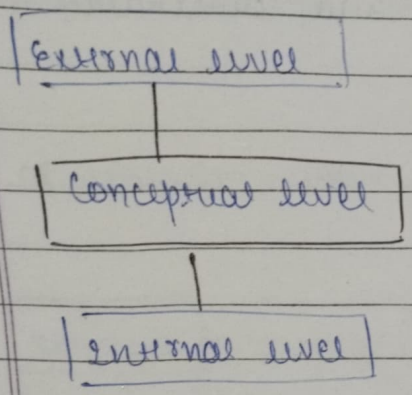
Section 'D'-

Que- Draw a simple ER diagram for any real life system and explain one key design decision.

- One customer can have multiple accounts.



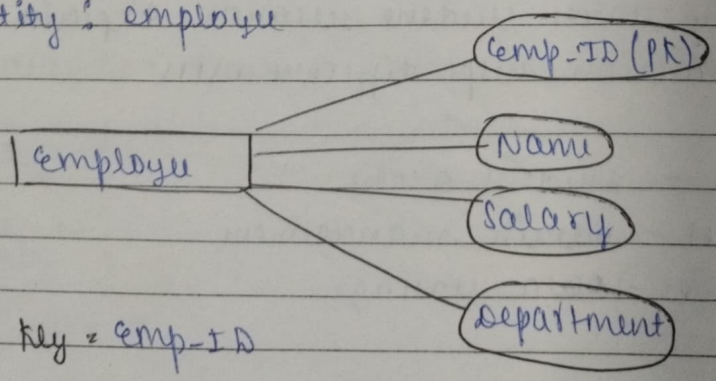
Que- Draw the three-schema architecture and maintain one benefit.



Benefit: provides data independence

Que- Draw an entity with attributes and identify its primary key.

- Entity: employee



- primary key = Emp-ID